

# Reproducible Processing of Home Blood-Pressure Time Series: Methodology and Operation of the Generator `generate_bp_tikz.py`

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**Abstract**—Home blood-pressure monitoring (HBPM) typically produces an irregularly sampled time series: several readings on some days, none on others. A naive average over all individual readings therefore over-weights measurement-intensive days and distorts the overall picture. This document describes the methodology and operation of `generate_bp_tikz.py`, a command-line tool that turns a simple CSV file into two coordinated LaTeX/PGFPlots diagrams and machine-readable statistics CSVs. The core principle is a two-stage aggregation: first by calendar day, then optionally into fixed-length blocks. We explain the computation variants (calendar-day-weighted mean versus median of daily medians), the outlier logic, and the generated data formats, and we document every parameter in tabular form. The goal is a reproducible, transparent, and clinically interpretable representation that explicitly shows comparison thresholds without implying aneurysm- or patient-specific target values.

**Index Terms**—Blood pressure, HBPM, time series, aggregation, median, interquartile range, LaTeX, PGFPlots, reproducibility.

## I. INTRODUCTION AND MOTIVATION

Self-measurement of blood pressure at home yields an irregularly sampled time series spanning weeks or months. Two methodological problems recur. First, the number of readings per day varies strongly; a day with eight readings would be weighted eight times as heavily as a day with a single reading in a naive overall average. Second, real HBPM data contain individual strongly deviating values (for example immediately after physical exertion) that affect robust and non-robust location measures differently.

`generate_bp_tikz.py` addresses both points through a deterministic, parameterizable pipeline. The input is a CSV file; the output consists of a fragment that can be included in an existing LaTeX document, two standalone documents, and two optional statistics CSVs. All location and dispersion measures are computed explicitly and traceably in the source, so the figures can be checked directly against the underlying numbers.

## II. DATA MODEL AND INPUT

A CSV file is expected with at least the columns *Date*, *Systolic*, and *Diastolic*; optional columns *Pulse*, *Time*, and *Note* are recognized and evaluated where useful. Column detection

uses alias names in German and English (e.g. *date/datum*, *systolic/systolisch*), with case and special characters normalized.

Dates are accepted in the formats `YYYY-MM-DD`, `DD.MM.YYYY`, `DD.MM.YY`, `DD/MM/YYYY`, and `DD/MM/YY`. Numeric values are parsed in both English and European notation: values such as 130, 130.5, 130,5, 1.234,56, and 1,234.56 are interpreted correctly by treating the last separator that occurs as the decimal separator. Non-numeric suffixes such as “mmHg” are tolerated. The input delimiter is auto-detected by default (comma, semicolon, or tab) and can be forced with `--delimiter`.

On input, only rows within the chosen date window (`--date-from` through optional `--date-to`) are accepted. Missing or invalid systolic or diastolic values cause a clear error abort with the row number, rather than silently producing corrupted statistics.

## III. AGGREGATION METHODOLOGY

### A. Stage 1: Aggregation per Calendar Day

In the first step, all readings of a calendar day are combined into a daily statistic. For systolic and diastolic pressure as well as for the derived pulse pressure (the difference between systolic and diastolic), the minimum, maximum, median, and arithmetic mean are determined; analogously for pulse, if present. This daily reduction is the decisive step against over-weighting measurement-intensive days: regardless of whether a day contains one or eight readings, it subsequently enters the further computations as *one* daily value.

The daily diagram (Figure 1 of the output) shows the daily median as a point and the daily range (minimum to maximum) as a vertical bar for each day. Days with only one reading therefore have no visible range. The median is preferred over the mean here because, within a day, it is robust against a single outlier and, together with the displayed range, captures both the typical level and the dispersion of the day.

### B. Stage 2: Block Aggregation

In the second step, the daily values are grouped into blocks of fixed length (default seven days, configurable

via `--block-days`). Block boundaries are counted from `--date-from`, so the first block starts independently of the weekday. The last block may be shorter; the caption explicitly notes when it comprises fewer days and is therefore less robust.

For each block, two distinct quantities are computed and deliberately displayed separately in the diagram:

- the *central line* as the block’s location measure, and
- the *interquartile range (IQR)* of the daily medians as a dispersion measure, drawn as an error bar from  $Q_1$  to  $Q_3$ .

The quartiles are determined by linear interpolation of the sorted daily medians, comparable to the default of common statistical packages.

#### IV. COMPUTATION VARIANTS OF THE CENTRAL LINE

The central methodological decision concerns the location measure of the block central line. The tool offers two variants via `--week-central`, each serving a clear purpose.

##### A. Mean (Default)

In the default (`mean`), the central line is the calendar-day-weighted mean of the daily means. Formally, for a block with  $D$  days and daily means  $\bar{x}_d$ :

$$\mu_{\text{block}} = \frac{1}{D} \sum_{d=1}^D \bar{x}_d. \quad (1)$$

Since  $\bar{x}_d$  is already the mean of the readings of day  $d$ ,  $\mu_{\text{block}}$  is a two-stage, day-equal-weighted mean. This variant is compatible with the clinically common HBPM/ESC mean thresholds, which are also based on means, and is therefore suitable for direct comparison with such thresholds.

##### B. Median of Daily Medians

With `median`, the central line is the median of the block’s daily medians:

$$m_{\text{block}} = \text{median}(\{\tilde{x}_d\}_{d=1}^D), \quad (2)$$

where  $\tilde{x}_d$  is the median of day  $d$ . This variant is more robust against individual outlier days and is also consistent with the displayed IQR box, since both are based on the same daily medians. It is recommended when the typical level should be shown despite occasional exertion spikes.

##### C. Recommendation

The choice is not a matter of right or wrong but of purpose: for comparison with mean-based thresholds, `mean` is natural; for an outlier-robust trend display, `median`. Because dispersion (IQR box) and location are shown separately, the information remains fully visible in both cases.

#### V. OUTLIER MARKING

Optionally (`--week-outliers`), the weekly diagram marks individual days whose daily median exceeds or falls below a clinical threshold as small points next to the respective IQR marker. The upper thresholds default to 135 mmHg systolic and 85 mmHg diastolic, matching the HBPM comparison lines drawn in the diagram; they are adjustable via `--week-outlier-sys-hi` and `--week-outlier-dia-hi`. Additionally, optional *lower* thresholds can be activated via `--week-outlier-sys-lo` and `--week-outlier-dia-lo`, for example to reveal days that are too low under therapy. This marking does not replace a statistical outlier definition but highlights clinically interesting individual days without removing them from the aggregation; all days remain included in mean, median, and IQR.

#### VI. REFERENCE ELEMENTS AND THRESHOLDS

Both diagrams contain the same reference elements, explained once in a shared paragraph and referenced by the captions. The light-gray corridors 120–129 mmHg systolic and 70–79 mmHg diastolic are general ESC orientations under therapy subject to individual tolerability and are explicitly *not* aneurysm-specific target lines. The dotted lines at 135 mmHg systolic and 85 mmHg diastolic mark frequently used home comparison thresholds. This deliberate separation between comparison threshold and individual therapy target prevents the figure from implying an unsupported, patient-specific target line.

The automatic summary line at the bottom of the daily diagram states the day-weighted mean, the median of daily medians, and the number of days with a daily mean below 135/85 mmHg. It can be suppressed with `--no-daily-summary-label`.

#### VII. GENERATED FILES

By default, the tool writes an includable LaTeX fragment (`--out`), a single-page standalone document with both diagrams stacked (`--standalone-out`), and a “two-sides” standalone document in which each diagram occupies its own tightly cropped page (`--two-sides-out`). Optionally, two statistics CSVs are written (`--daily-stats`, `--weekly-stats`). The single-page document scales the diagram heights at typesetting time from the available page height, so the page is always fully used regardless of the length of the captions.

Tables I and II document the columns of the two statistics CSVs.

#### VIII. ANONYMIZED APPLICATION EXAMPLE

Figures 1 and 2 show the two generated diagrams on an anonymized, realistic data set (101 readings on 37 calendar days). The figure labels are produced by the tool in German; an English gloss is given in the captions. Figure 1 is the daily course with daily median, daily range, the automatic

TABLE I: Columns of the daily-statistics CSV (--daily-stats).

| Column           | Meaning                                |
|------------------|----------------------------------------|
| date             | calendar day (ISO 8601)                |
| day_index        | day index from date-from (0-based)     |
| n                | number of readings on this day         |
| sys_min/max      | systolic minimum/maximum               |
| sys_median       | systolic daily median                  |
| sys_mean         | systolic daily mean                    |
| dia_min/max      | diastolic minimum/maximum              |
| dia_median       | diastolic daily median                 |
| dia_mean         | diastolic daily mean                   |
| pulse_*          | pulse min/max/median/mean (if present) |
| pulse_pressure_* | pulse pressure min/max/median/mean     |

TABLE II: Columns of the block-statistics CSV (--weekly-stats).

| Column                  | Meaning                                |
|-------------------------|----------------------------------------|
| block                   | block index (0-based)                  |
| start_date, end_date    | block start/end                        |
| x_mid                   | x position (day index) of block center |
| n_days                  | number of days in the block            |
| n_readings              | number of readings in the block        |
| label                   | block axis label                       |
| sys_mean_of_daily_means | weighted systolic mean                 |
| sys_q1/q3_daily_medians | systolic $Q_1/Q_3$ of daily medians    |
| dia_mean_of_daily_means | weighted diastolic mean                |
| dia_q1/q3_daily_medians | diastolic $Q_1/Q_3$ of daily medians   |
| pulse_pressure_*        | analogous for pulse pressure           |

summary line, and the  $n=$  labels (readings per day). Figure 2 is the condensed 7-day overview; here the median central line (--week-central median) and outlier marking (--week-outliers) were chosen. In both figures the ESC corridors and HBPM comparison lines are drawn as a shared reference.

Figure 3 additionally shows the single-page standalone document (--standalone-out), in which both diagrams, the shared reference paragraph, and the captions are combined on one page. The diagram heights are computed at typesetting time from the available page height, so the page is fully used regardless of the caption length.

Finally, Figure 4 shows the two-sides variant (--two-sides-out), in which each diagram occupies its own tightly cropped page and the corresponding caption is set as a text node below the axis. This format is particularly suited to embedding the individual diagrams as separate appendix figures.

## IX. PARAMETER REFERENCE

Table III lists all command-line parameters with default and purpose.

## X. OPERATION

### A. Prerequisites

The including LaTeX document requires:

```
\usepackage{tikz}
\usepackage{pgfplots}
\pgfplotsset{compat=1.18}
```

The script itself has no external Python dependencies beyond the standard library.

### B. Minimal Call

```
python3 generate_bp_tikz.py \
--csv bp_raw.csv \
--date-from 2026-05-15
```

Produces the fragment and both standalone documents with default names.

### C. With Date Window and Statistics Export

```
python3 generate_bp_tikz.py \
--csv bp_raw.csv \
--date-from 15.05.2026 \
--date-to 20.06.2026 \
--out bp_fragment.tex \
--daily-stats bp_daily_stats.csv \
--weekly-stats bp_weekly_stats.csv
```

### D. Median Central Line with Outliers

```
python3 generate_bp_tikz.py \
--csv bp_raw.csv \
--date-from 2026-05-15 \
--week-central median \
--week-outliers
```

### E. Fully Specified Call

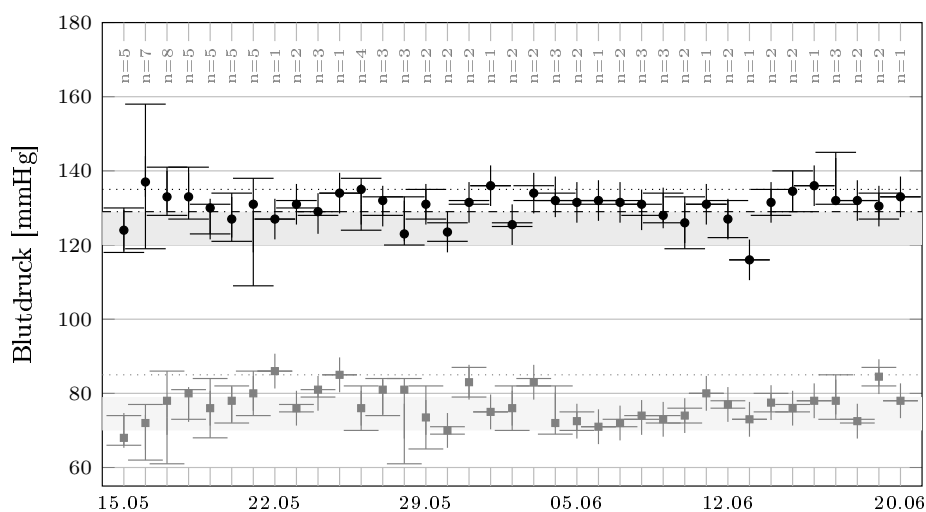
```
python3 generate_bp_tikz.py \
--csv bp_raw.csv \
--date-from 2026-05-15 \
--date-to 2026-06-20 \
--week-central median --week-outliers \
--week-outlier-sys-hi 135 \
--week-outlier-dia-hi 85 \
--show-daily-n \
--out bp_fragment.tex \
--standalone-out bp_onepage.tex \
--two-sides-out bp_two_sides.tex \
--two-sides-width-cm 16 \
--daily-stats bp_daily_stats.csv \
--weekly-stats bp_weekly_stats.csv
```

### F. Including the Output

The fragment is included in running text with `\input{bp_fragment.tex}`. Alternatively, the single-page or two-sides standalone can be compiled separately and the resulting PDF embedded as an appendix figure.

## XI. REPRODUCIBILITY AND VALIDATION

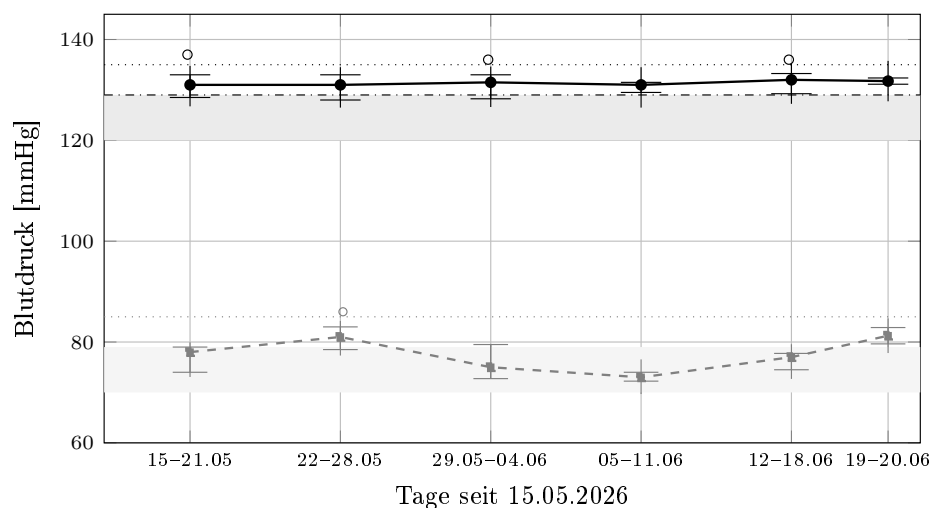
Since all location and dispersion measures are computed explicitly in the source and output via the statistics CSVs, the figures can be checked directly against the numbers. A typical validation step is to compare the summary line of the daily diagram (weighted mean, median of daily medians, number of days below 135/85 mmHg) with the values in the daily-statistics CSV. The same input, the same date window, and the same parameters lead deterministically to identical output.



Tage seit 15.05.2026 | Mittelwert 130.4/76.6 mmHg | Median 131.5/76.0 mmHg | <135/85: 31/37 Tage

— ESC-Zielkorridor syst. 120–129 — ESC-Zielkorridor diast. 70–79 - - - Obergrenze ESC-Korridor syst. 129  
 | Systolische Tages-Spannweite • Systolischer Tagesmedian | Diastolische Tages-Spannweite  
 ■ Diastolischer Tagesmedian ..... HBPM-Vergleich syst. 135 ..... HBPM-Vergleich diast. 85

Fig. 1: Anonymized example, daily course. Points show the daily median, vertical bars the daily range (min–max); gray n= labels at the top give the readings per day. Summary line glosses: *Tage seit* = days since; *Mittelwert* = mean; *Median*; *Tage* = days; *Blutdruck* = blood pressure.



— ESC-Zielkorridor syst. 120–129 — ESC-Zielkorridor diast. 70–79 - - - Obergrenze ESC-Korridor syst. 129  
 | Systolischer IQR der Tagesmediane — Systolischer 7-Tage-Median | Diastolischer IQR der Tagesmediane  
 - - - Diastolischer 7-Tage-Median ○ Systolische Ausreißer-Tage ○ Diastolische Ausreißer-Tage  
 ..... HBPM-Vergleich syst. 135 ..... HBPM-Vergleich diast. 85

Fig. 2: Anonymized example, 7-day overview with median central line and outlier marking. Lines show the median of daily medians, error bars the interquartile range of daily medians; circles mark individual days whose daily median exceeds the threshold (135/85 mmHg). Legend glosses: *IQR der Tagesmediane* = IQR of daily medians; *7-Tage-Median* = 7-day median; *Ausreißer-Tage* = outlier days; *HBPM-Vergleich* = HBPM comparison.

TABLE III: Complete parameter reference of `generate_bp_tikz.py`.

| Parameter                             | Default                                              | Purpose                                                                 |
|---------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------|
| <code>--csv</code>                    | — (required)                                         | Input CSV with date, systolic, diastolic (optional time/pulse/note).    |
| <code>--date-from</code>              | — (required)                                         | First date in the evaluation window.                                    |
| <code>--date-to</code>                | all from date-from                                   | Last date; if omitted, all subsequent values.                           |
| <code>--out</code>                    | <code>bp_diagrams.tex</code>                         | Path of the includable LaTeX fragment.                                  |
| <code>--daily-stats</code>            | off                                                  | Optional daily statistics as CSV.                                       |
| <code>--weekly-stats</code>           | off                                                  | Optional block statistics as CSV.                                       |
| <code>--block-days</code>             | 7                                                    | Block length of the second aggregation stage, in days.                  |
| <code>--delimiter</code>              | auto                                                 | Delimiter: auto/comma/semicolon/tab.                                    |
| <code>--standalone-out</code>         | <code>bp_diagrams_both_onepage_standalone.tex</code> | Single-page standalone with both diagrams.                              |
| <code>--no-standalone</code>          | off                                                  | Do not write the single-page standalone.                                |
| <code>--standalone-title</code>       | <code>Blutdruckdiagramme</code>                      | Title above the standalone diagrams.                                    |
| <code>--show-daily-n</code>           | off                                                  | Gray $n=$ labels (readings per day) in the daily diagram.               |
| <code>--daily-n-y</code>              | automatic                                            | Fixed $y$ position of the $n=$ labels.                                  |
| <code>--week-outliers</code>          | off                                                  | Mark outlier days in the weekly diagram.                                |
| <code>--week-outlier-sys-hi</code>    | 135.0                                                | Upper systolic outlier threshold (mmHg).                                |
| <code>--week-outlier-dia-hi</code>    | 85.0                                                 | Upper diastolic outlier threshold (mmHg).                               |
| <code>--week-outlier-sys-lo</code>    | off                                                  | Optional lower systolic threshold.                                      |
| <code>--week-outlier-dia-lo</code>    | off                                                  | Optional lower diastolic threshold.                                     |
| <code>--week-central</code>           | mean                                                 | Central line: mean (weighted mean) or median (median of daily medians). |
| <code>--no-daily-summary-label</code> | off                                                  | Suppress the automatic summary line in the daily diagram.               |
| <code>--two-sides-out</code>          | <code>bp_diagrams_standalone_two_sides.tex</code>    | Two-sides standalone (one cropped page per diagram).                    |
| <code>--no-two-sides</code>           | off                                                  | Do not write the two-sides standalone.                                  |
| <code>--two-sides-width-cm</code>     | 16.0                                                 | Fixed axis width (cm) of the two-sides variant.                         |

## XII. IMPLEMENTATION AND TESTED ENVIRONMENT

The tool is implemented in Python 3 and uses only the standard library (no third-party dependencies); Python 3.9 or newer is recommended. The generated documents were typeset with `pdflatex` (TeX Live) using `tikz` and `pgfplots` (`compat=1.18`); the accompanying IEEE papers use `IEEEtran.cls` version V1.8b.

Testing was performed *only* on the following platforms: (i) macOS with desktop Python and a desktop TeX Live/MacTeX installation; (ii) iPad and iPhone with *a-Shell*, a local iOS terminal that bundles Python 3 and TeX Live (with TikZ and LuaTeX) – both the script and the LaTeX compilation run on-device; and (iii) *Texifier* (formerly TeXpad) on macOS and iPadOS/iOS for editing and typesetting the LaTeX documents. Note that Texifier’s built-in live typesetter (TexpadTeX) runs in a sandbox and does not permit shell-escape (`\write18`); the Python script itself is therefore run in *a-Shell* or an ordinary shell, while Texifier is used to typeset the generated `.tex` files. Other platforms (Windows, Linux) are expected to work with a standard Python 3 and TeX Live installation but have not been tested.

## XIII. LIMITATIONS

The tool is a processing and visualization tool, not a diagnostic system. The thresholds drawn are comparison and orientation lines; the determination of individual target values

rests with the treating physician. The expressiveness of short blocks with few days is limited and is marked accordingly in the caption. The outlier marking is a threshold-based highlighting and not a statistical outlier classification.

## XIV. SUMMARY

`generate_bp_tikz.py` converts irregularly sampled HBPM time series via a two-stage aggregation into two coordinated, reproducible diagrams. The separate display of location (mean or median) and dispersion (range or IQR), the transparent outlier logic, and the explicit separation of comparison threshold and therapy target enable a clinically interpretable yet traceable representation. Full parameterization via the command line makes the tool suitable both for one-off evaluations and for recurring, automated reports.

## ACKNOWLEDGMENT AND DISCLOSURE OF GENERATIVE AI USE

In accordance with the IEEE guidelines on the use of AI-generated content, the following is disclosed. The AI system Claude (Anthropic) was used in the preparation of this document. The AI was used in an assistive capacity for: the linguistic drafting and structuring of the running text; the formulation of the methodology description based on the source code developed by the author; the creation of the LaTeX and table structure; and the preparation of the anonymized example figures. The conception, software, underlying data, technical

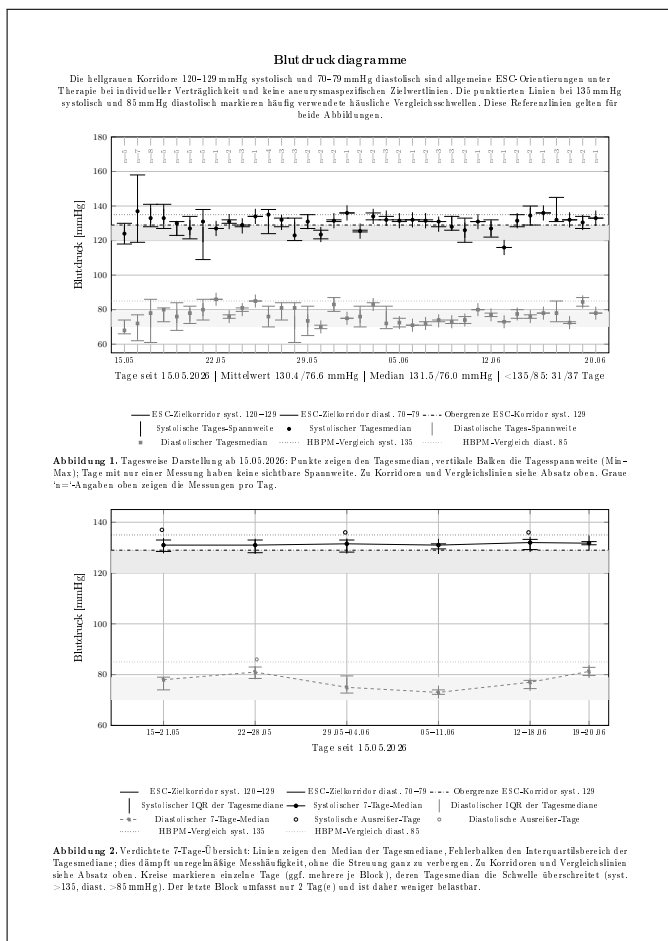
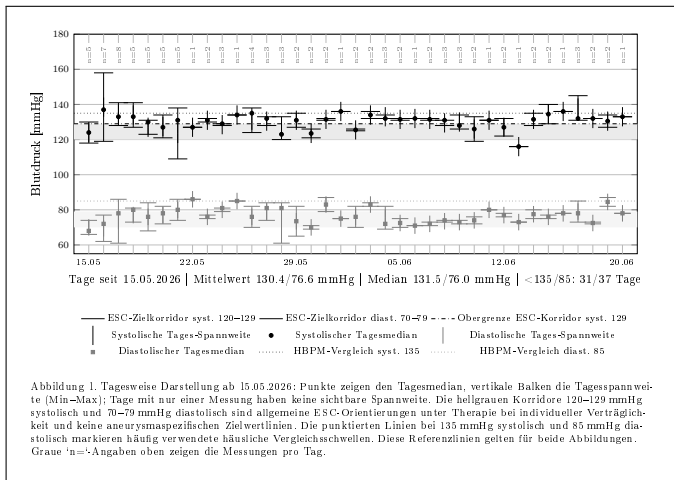
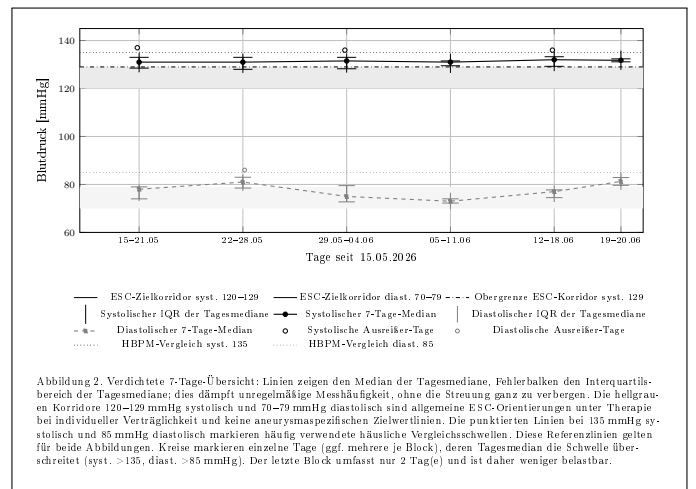


Fig. 3: Anonymized example of the single-page standalone document (`--standalone-out`): both diagrams, the shared reference paragraph, and the captions on one page, with the diagrams scaled to the page height. The figure labels are produced by the tool in German; see Figures 1 and 2 for English glosses of the recurring terms.

claims, and final responsibility for the content rest entirely with the author; all AI-assisted contributions were reviewed, verified, and corrected where necessary by the author. The AI is not an author of this work. A more detailed description of the AI contribution is provided in the file `AI_USAGE.md` of the accompanying repository.



(a) Page 1: daily course.



(b) Page 2: 7-day overview.

Fig. 4: Anonymized example of the two-sides variant (`--two-sides-out`): each diagram occupies its own cropped page with the caption set as a node below the axis. The figure labels are produced by the tool in German; see Figures 1 and 2 for English glosses.